

Yiming (Emmett) Peng

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EDUCATION

- **University of Toronto** Sept 2026 (Expected)
Doctor of Philosophy in Biostatistics
Research interests: statistical machine learning, causal inference, and representation learning
◦ Advisors: Kuan Liu, Zihang Lu
- **University of Toronto** Sept 2024 – Aug 2026 (Expected)
Master of Science in Biostatistics – Thesis Option; GPA: 4.00/4.00
Thesis: Bayesian Nonparametric Latent Class Framework for Causal Inference in Cognitive Trajectories
◦ Advisors: Kuan Liu, Zihang Lu
- **Beijing Institute of Technology, Zhuhai** Sept 2020 – Jun 2024
Bachelor of Science in Applied Statistics; GPA: 3.96/4.00 (WES)
Thesis: Deep Learning and Stochastic Modeling of Human Activity Patterns through Smartwatch Sensors for Health Interventions (*Honored as Outstanding Thesis*)
◦ Advisor: Han C.W. Hsiao

PUBLICATIONS AND MANUSCRIPTS

R=UNDER REVIEW, S=IN SUBMISSION, W=WORKING PAPERS

- [R] Denier, D., Peng, Y. (Emmett), ..., Ricciuto, A. (2026) **Longitudinal GGT Trajectories Identify Prognostic Phenotypes in Pediatric Primary Sclerosing Cholangitis**. *Under review at JHEP Reports*.
- [S] Peng, Y. (Emmett)*, Truong, V.*, Xie, A., Shi, Y., Hu, P. (2026) **PRECISE: A Framework for the Prediction of Response Using Cell-Type Inference and Single-Cell Embedding in Breast Cancer Patients Receiving Anti-PD-1 Treatment**. *In Submission*. [\[Code\]](#)
- [W] Peng, Y. (Emmett), Lu, Z., Liu, K. (2026) **Bayesian Nonparametric Latent Class Framework for Causal Inference in Cognitive Trajectories**. *Manuscript in preparation*.
- [W] Zhang, J., Shi, Y., Xie, A., Peng, Y. (Emmett), Hu, P. (2026) **Predicting New-Onset Atrial Fibrillation with Competing Risks Using an Uncertainty-Calibrated Interpretable Tabular Transformer: Evidence from the Cardiovascular Imaging Registry of Calgary (CIROC)**. *Manuscript in preparation*.
- [W] Khan, A., Peng, Y. (Emmett), Zhang, I., Chen, H. (2026) **Robust and Interpretable Machine Learning for Atypical Parkinson's Disease Classification Under Distribution Shift**. *Working paper*.
- [W] Jing, Y., Peng, Y. (Emmett), Danguedan, A., Knight, A. (2026) **Examining the Relationship Between Socioeconomic Factors and Mental Health in Childhood-Onset Systemic Lupus Erythematosus Using a Multilevel Analysis**. *Working paper*.

PRESENTATIONS

- [1] Peng, Y. (Emmett), Lu, Z., Liu, K. (2026) **Bayesian Nonparametric Latent Class Framework for Causal Inference in Cognitive Trajectories**. Accepted presentations at the *Statistical Society of Canada (SSC) Annual Meeting, Joint Statistical Meetings (JSM)*, and *ICSA-Canada Chapter Symposium*, 2026. *Forthcoming*.
- [2] Peng, Y. (Emmett), Denier, D., ..., Ricciuto, A. (2026) **Longitudinal GGT Trajectories Identify Prognostic Phenotypes in Pediatric Primary Sclerosing Cholangitis**. Poster presentation, *Canadian Statistics Student Conference (CSSC)*, Hamilton, Ontario, Canada, May 2026. *Forthcoming*.
- [3] Peng, Y. (Emmett), Truong, V., Xie, A., Shi, Y., Hu, P. (2025) **Enhancing Breast Cancer Treatment Response Prediction with Single-Cell RNA Sequencing and Large Language Models**. Oral presentation and poster, *2025 Statistical Society of Canada (SSC) Annual Meeting*, p. 58. May 25–28, Saskatoon, Saskatchewan, Canada. [\[Slides\]](#) [\[Poster\]](#)
- [4] Zhang, J., Shi, Y., Xie, A., Peng, Y. (Emmett), Hu, P. (2025) **Uncertainty-Calibrated Interpretable Tabular Transformer Model for Atrial Fibrillation Prediction with Competing Risk**. Poster, *2025 Statistical Society of Canada (SSC) Annual Meeting*, p. 42. May 25–28, Saskatoon, Saskatchewan, Canada. [\[Poster\]](#)
- [5] Peng, Y. (Emmett), Lu, Z., Liu, K. (2025) **Bayesian Latent Class Causal Inference with Nonparametric Covariate Modeling for Endpoint Cognitive Outcomes in Aging Populations**. Invited poster presentation, *Data Science Institute Causal Inference Workshop*, Toronto, Ontario, Canada, October 22, 2025. [\[Poster\]](#)

PROFESSIONAL EXPERIENCE

- **The Hospital for Sick Children**
Research Volunteer, supervised by Dr. Amanda Ricciuto and Dr. Anne Griffiths
 - Implement latent class mixed models in R to analyze longitudinal serum biomarker trajectories in primary sclerosing cholangitis (PSC).
 - Integrate latent class membership into survival analysis to assess prediction of transplant outcomes, and collaborate with clinicians and biostatisticians on interpretation and manuscript preparation.
 - **Knight Lab, The Hospital for Sick Children**
Clinical Research Project Assistant, supervised by Dr. Andrea Knight
 - Implement Bayesian statistical methods in R to study socioeconomic and mental health outcomes in childhood-onset systemic lupus erythematosus (cSLE).
 - Lead statistical analyses, including data pre-processing, imputation, and modeling, collaborate on study design, and provide statistical consulting while contributing to manuscript preparation.
 - **Hu Lab, Dalla Lana School of Public Health, University of Toronto**
Data Science Research Student, supervised by Dr. Pingzhao Hu
 - Developed an integrated multi-stage pipeline in Python combining single-cell RNA-seq data with embeddings from large language models (LLMs) and foundation models to identify cell type-specific biomarkers for breast cancer treatment response.
 - Implemented supervised classification using cell-type-specific biomarkers and cluster-specific bulk gene-expression profiles to predict treatment response, achieving improved predictive performance over baseline.
 - **The University of Hong Kong**
Research Volunteer, supervised by Dr. Liwu Zheng
 - Collaborate with clinicians on biostatistical research projects.
 - Provide statistical consulting for clinical studies.

Jun 2025 - Present

Toronto, Canada

Jun 2025 - Mar 2026

Toronto, Canada

Oct 2024 - Aug 2025

Toronto, Canada

Jan 2025 - Apr 2025
- HONORS AND AWARDS
- Amounts in Canadian dollars unless otherwise specified

- **ICSA-Canada Chapter Symposium Student/Postdoc Travel Scholarship Award - \$600**
ICSA-Canada Chapter Symposium
 - **Winner – Case Studies in Data Analysis Competition, SSC 2025 - \$2,500 per team**
Statistical Society of Canada (SSC)
 - **CSSC 2025 Travel Award - \$150**
Canadian Statistics Student Conference (CSSC), sponsored by CANSSI
 - **SGS Conference Grant - \$450**
University of Toronto
 - **BITZH President’s Scholarship - \$2,000**
Beijing Institute of Technology, Zhuhai
 - **College Dean’s Honor Scholarship - \$7,000**
Beijing Institute of Technology, Zhuhai
 - **BITZH Freshman Entrance Scholarship and Continuing Scholarship - \$18,000**
Beijing Institute of Technology, Zhuhai
 - **BITZH Outstanding Student Scholarship**
Beijing Institute of Technology, Zhuhai

2026

2025

2025

2025

2023

2021 - 2024

2020 - 2024

2020 - 2024

SKILLS

- **Programming Languages:** R, Python, MATLAB, C++
 - **Tools & Technologies:** Git, Linux, HPC (Compute Canada, Slurm), LaTeX, RMarkdown, Jupyter Notebook
 - **Statistical Methods:** Causal Inference, Latent Class Analysis, Survival Analysis, Longitudinal Data Analysis
 - **Machine Learning Models:** Transformers, RNNs, Regression (GLM, Penalization), CNNs, Neural Networks
 - **Languages:** Chinese (native), English (professional working proficiency)

COMMUNITY INVOLVEMENT AND LEADERSHIP EXPERIENCE

- **ASA DataFest at UofT**
Judge and Mentor
 - **Biostatistics Union of Graduate Students (BUGS)**
PHSA MSc Representative and Seminar Committee member
 - **Health Data Working Group**
Student Member
 - **Public Health Students’ Association, University of Toronto**
Biostatistics MSc Representative
 - **Student Union, College of Global Talent, Beijing Institute of Technology, Zhuhai**
Officer

May 2026

Sep 2024 - Present

Sep 2024 - Present

Sep 2025 - Present

Sep 2020 – Jun 2022





